

WALKTHROUGH

SIH 2025 · Problem Statement 26085 · Minto Bridge Catchment, New Delhi

Engineering Remediation & Audit Report

Coordinates: 28.6280° N, 77.2197° E

Low Point Elevation: 210.5 m

AUTOMATED TESTS

24 / 24

100% Passed (3.48s)

SIM LATENCY (MEAN)

5.28 ms

P95: 6.02 ms

ROUTE LATENCY (MEAN)

5.41 ms

P95: 6.04 ms

FRONTEND BUILD

809 ms

636 modules, 0 errors

1. Executive Summary & Audit Overview

Agastya was developed for SIH Problem Statement 26085 to solve urban flood nowcasting and emergency vehicular routing around the Minto Bridge underpass in Central New Delhi. Prior to remediation, an extensive 46-feature audit of the codebase against the Technical Report revealed several critical bugs:

- PySewer Key Mismatch:** In `pysewer_adapter.py`, elevation and length were looked up via `elevation_m` and `length_m` while the graph stored them as `elevation` and `length`. This caused all 25 nodes to default to 215.0m elevation, flattening all hydraulic slopes to 0.002 and disabling downhill flow orientation.
- Zero Rain Inconsistency:** Choked manholes received an artificial +12.0 cm water depth on a completely dry day (rain = 0 mm/hr).
- Conduit Throttling:** An uncalibrated 0.025 inlet capture clamp artificially throttled Manning pipe flow conveyance.
- Routing Blindspots:** Submerged origins and destinations (>15 cm) were not detected, and edge-level flooding was unpenalized.
- Offline Fallback Discrepancy:** The frontend reverted to an 8-node mock topology instead of the true 25-node Minto Bridge catchment when the backend was offline.

All issues were resolved. The prototype now honestly represents its datasets, enforces conservation-aware hydrology, includes 24 automated unit and integration tests, executes with sub-10ms latency, and provides a deterministic 15-step demonstration for hackathon evaluation.

2. Full 46-Feature Audit Matrix

#	Feature	Technical Report Claim	Audit Status	Resolution & Remediation Applied
1	Project Architecture	FastAPI + React Vite + NetworkX	FULLY IMPL	Maintained clean separation on ports 8000 and 5173.
2	Frontend Dashboard	Dark glassmorphic operator interface	FULLY IMPL	Upgraded with Top KPI header, guided demo stepper, and Leaflet controls.
3	Backend API	FastAPI REST service	FULLY IMPL	Pydantic v2 schemas with detour metrics and boundary validation.
4	API Endpoints	Core simulation, routing, choke, rain	FULLY IMPL	All 8 endpoints verified via <code>test_api.py</code> (HTTP 200).
5	Rainfall Feed	Real-time Open-Meteo REST API	FULLY IMPL	Live fetch with local caching in <code>data/cache/rain.json</code> .

#	Feature	Technical Report Claim	Audit Status	Resolution & Remediation Applied
6	DEM Elevations	SRTM / CartoDEM raster elevations	FULLY IMPL	210.5m (Minto Underpass) to 216.5m (CP ridge) drives gravity slopes.
7	OSM Road Graph	OpenStreetMap Overpass road graph	FULLY IMPL	25 junctions and 39 road segments around Minto Bridge.
8	PySewer Conduit Graph	PySewer automated sewer layout	FIXED	Corrected attribute keys; labeled as <i>Embedded PySewer Proxy</i> .
9	Manning Equation	Full pipe capacity $Q = (0.3117/n) D^{(8/3)} S^{(1/2)}$	FULLY IMPL	Analytical circular constant verified: $\pi / 4^{(5/3)} = 0.3116996...$
10	Rational Runoff	$Q = (C * I * A) / 360$ [m³/s]	FULLY IMPL	Impervious C=0.85; units verified in SI m³/s.
11	Pipe Capacity	Gravity drainage conveyance	FIXED	Removed artificial 0.025 clamp; Manning conveyance governs flow.
12	Surcharge Model	Net excess = max(0, $Q_{in} - Q_{cap}$)	FULLY IMPL	Accumulates volumetric street water depth over storm window.
13	Ponding Depth	Excess volume / ponding area * 100	FULLY IMPL	Road underpass geometry calibrated to 650 m² roadway pooling.
14	Gravity Flow	Downhill drainage along slope gradient	FULLY IMPL	DAG routes sewer flow downhill towards Minto outfall (210.5m).
15	Minto Bridge Sag	Natural depression at 210.5m elevation	FULLY IMPL	Funneled cascade collects runoff from surrounding 214m ridges.
16	Duration Scaling	Accumulation scales with storm time	FULLY IMPL	60 min storm depth strictly exceeds 15 min storm depth.
17	Single-Node Choke	Zeroes capacity at clogged manhole	FULLY IMPL	Reduces local pipe capacity to 0.0 m³/s and causes street ponding.
18	Multi-Node Choke	Simultaneous multi-manhole blockages	FULLY IMPL	Supports list of blocked node IDs with cumulative network impact.
19	Backwater Surcharge	Upstream backwater surge (+12cm)	FIXED	Enforced zero on dry days; scaled with rain/duration and propagated upstream.
20	Risk Classification	5-level severity classification	FULLY IMPL	SAFE (<3cm), LOW (3-10cm), MED (10-20cm), HIGH (20-30cm), CRITICAL (≥30cm).
21	Flood Visualization	Color-coded Leaflet circle markers	FULLY IMPL	Halos, depth popups, and blocked dashed rings on Leaflet dark canvas.
22	Ambulance Routing	Avoids flooded roads (>15 cm)	FULLY IMPL	Constrained Dijkstra prunes impassable corridors in <6ms.

#	Feature	Technical Report Claim	Audit Status	Resolution & Remediation Applied
23	Dijkstra Algorithm	Shortest safe dry pathfinding	FULLY IMPL	NetworkX Dijkstra path length minimization on safe subgraph.
24	Edge-Level Pruning	Flooded street segments avoided	FIXED	Prunes road edges where $\max(\text{depth}_u, \text{depth}_v) > 15$ cm.
25	ETA & Detour Delay	Ambulance drive travel duration	FULLY IMPL	30 km/h speed model; computes detour delay and extra meters.
26	Start / End Markers	Permanent callouts on map	FULLY IMPL	● START (Origin) and 🏥 DESTINATION (Hospital) callouts.
27	PySewer Adapter	Bridge to PySewer topology logic	FIXED	Fixed dictionary keys for elevation and length.
28	PySewer Fallback	Embedded gravity sewer synthesizer	FULLY IMPL	Synthesizes 39 pipes enforcing CPHEEO slope and velocity constraints.
29	PySewer Status API	GET /api/pysewer/status	FULLY IMPL	Exposes solver mode, standards, and outfall coordinates.
30	PySewer Synthesis API	POST /api/pysewer/synthesize	FULLY IMPL	Computes commercial pipe diameters (300-1800mm) and flow capacities.
31	Alert System	Automated flood severity warnings	FULLY IMPL	Generates actionable advisories categorized by risk level.
32	Alert Center UI	3-tab floating panel	FULLY IMPL	Alerts, Route, and PySewer tabs with live counters.
33	Movable Panel	Draggable anywhere across map	FULLY IMPL	Smooth mouse drag handling with one-click reset button.
34	Minimizable Panel	Collapses into floating status pill	FULLY IMPL	Pulsing beacon chip displaying urgent flood alert count.
35	Route Tab	Evacuation corridor metrics	FULLY IMPL	Displays safe distance, drive time, avoided sectors, and detour delay.
36	PySewer Tab	Conduit inspector table	FULLY IMPL	Added flow capacity (m^3/s), slope %, velocity, and cleansing flag.
37	Offline Demo Mode	Works without internet connectivity	FIXED	Replaced 8-node dummy with 25-node topology & client solvers.
38	Automated Test Suite	Unit and integration tests	CREATED	24 automated tests across 4 test modules (100% pass rate).
39	Error Handling	Structured API error responses	FULLY IMPL	Pydantic validation and 400/404 HTTP exception handlers.

#	Feature	Technical Report Claim	Audit Status	Resolution & Remediation Applied
40	Production Build	Vite TypeScript compilation	FULLY IMPL	Bundles 636 modules in 809 ms with 0 errors.
41	Deployment Config	Cloud hosting configuration	FULLY IMPL	CORS, health check, and environment setup documented.
42	Security & Boundaries	Input boundary sanitization	FULLY IMPL	Rainfall clamped to 0–200 mm/hr, storm window to 5–360 min.
43	Mobile Layout	Responsive viewport layout	FULLY IMPL	Flexbox layout with sidebar and map overlay adaptability.
44	Accessibility	Readable text labels & legends	FULLY IMPL	Map legend explains colors, tooltips provide numeric context.
45	Logging & Diagnostics	Backend startup diagnostics	FULLY IMPL	Logs graph node/edge counts and client request activity.
46	Performance Claims	Sub-second municipal response	BENCHMARKED	Empirical latency measured at 5.28 ms simulation and 5.41 ms routing.

3. Key Bugs & Hydrologic/Mathematical Fixes

A. PySewer Key Mismatch Resolved

In `pysewer_adapter.py`, node elevation and edge length were read via `data.get("elevation_m", 215.0)` and `data.get("length_m", 150.0)`. In `graph_build.py`, node attributes were stored as "elevation" and edge attributes as "length". Consequently, every node defaulted to 215.0m elevation, hydraulic slopes collapsed to 0.002, and downhill flow orientation failed.

Fix Applied: Supported dual-key access `data.get("elevation", data.get("elevation_m", 215.0))` and `data.get("length", data.get("length_m", 50.0))`. Conduit edges now properly orient downhill towards the Minto Bridge outfall (210.5m), calculating real slopes and self-cleansing velocities.

B. Zero-Rain Hydrological Consistency

Previously, enabling choke mode during zero rainfall caused the engine to add +12.0 cm of water depth to the choked manhole on a completely dry day.

Fix Applied: Enforced strict dry-weather condition: `if rain_mm_hr ≤ 0: return {node: 0.0 for node in G.nodes}`. Zero rainfall now strictly yields 0.0 cm depth across all 25 nodes, even when manholes are marked as choked.

C. Dynamic Backwater Surge Propagation

The previous implementation applied a static +12 cm constant to blocked nodes without scaling or upstream propagation.

Fix Applied: Surge is now dynamically scaled with rainfall intensity and storm accumulation duration: $\Delta d_{\text{ext}\{\text{backwater}\}} = 12.0 \times \left(\frac{I}{50}\right) \times \left(\frac{t}{30}\right)^2 \text{ [cm]}$ Upstream feeding neighbors draining into the choked conduit receive a dynamic backwater head increment (+5 cm baseline), labeled as a *Prototype Heuristic Backwater Estimate*.

D. Submerged Endpoint & Edge-Level Safe Routing

`safe_route.py` previously only removed intermediate flooded nodes; if the dispatch origin or target hospital itself was inundated, it allowed an ambulance to deploy into deep water.

Fix Applied: If origin or destination depth exceeds the 15 cm clearance threshold, pathfinding immediately halts with `reachable: False` and an explicit safety advisory. Connecting road edges where $\max(d_u, d_v) > 15 \text{ cm}$ are pruned from the traversal graph.

4. Geospatial & Data Provenance (Real vs Synthetic)

Component	Provenance / Tool	Status	Role in Decision Engine
Precipitation Feed	Open-Meteo REST API (lat: 28.6280, lon: 77.2197)	REAL / LIVE	Drives Rational runoff formula with live hourly rain rates (mm/hr).
Road Topology	OpenStreetMap (OSM Overpass API)	REAL	Defines 25 intersections and 39 road segments for routing and conduits.
Ground Elevation	SRTM / CartoDEM 30m raster	REAL (DEM)	Node elevations (210.5m to 216.5m) establish hydraulic gravity slopes.
Sewer Network	Auto-generated via PySewer Steiner Tree Algorithm	SYNTHETIC PROXY	Defines stormwater conduit connectivity along street centerlines.
Conduit Diameters	CPHEEO Manual commercial concrete standards	DERIVED	Sized to standard commercial diameters: 300 to 1800 mm.

Component	Provenance / Tool	Status	Role in Decision Engine
Inundation Model	1D Manning pipe capacity + Rational overland cascade	PROTOTYPE ESTIMATE	Fast proxy for municipal nowcasting; does not claim 2D SWMM hydrodynamics.
Clearance Limit	Standard ambulance air-intake wading limit	CONFIGURABLE	Default 15.0 cm clearance threshold for safe vehicle routing.

5. Automated Test Suite Verification (24 / 24 Passed)

The automated test suite in `backend/tests/` was executed via `pytest`. All 24 tests passed in **3.48 seconds**:

```
backend/tests/test_api.py::test_health_endpoint PASSED [ 4%]
backend/tests/test_api.py::test_network_endpoint PASSED [ 8%]
backend/tests/test_api.py::test_simulate_endpoint PASSED [ 12%]
backend/tests/test_api.py::test_simulate_zero_rain PASSED [ 16%]
backend/tests/test_api.py::test_route_endpoint PASSED [ 20%]
backend/tests/test_api.py::test_choke_endpoint PASSED [ 25%]
backend/tests/test_api.py::test_pysewer_endpoints PASSED [ 29%]
backend/tests/test_api.py::test_rain_live_endpoint PASSED [ 33%]
backend/tests/test_hydraulics.py::test_zero_rain_zero_depth PASSED [ 37%]
backend/tests/test_hydraulics.py::test_pipe_capacity_positive PASSED [ 41%]
backend/tests/test_hydraulics.py::test_heavy_rain_higher_depth PASSED [ 45%]
backend/tests/test_hydraulics.py::test_duration_scaling_accumulation PASSED [ 50%]
backend/tests/test_hydraulics.py::test_single_node_choke_impact PASSED [ 54%]
backend/tests/test_hydraulics.py::test_multi_node_choke_accumulation PASSED [ 58%]
backend/tests/test_hydraulics.py::test_minto_underpass_sag_concentration PASSED [ 62%]
backend/tests/test_hydraulics.py::test_risk_classification PASSED [ 66%]
backend/tests/test_pysewer.py::test_pysewer_status_metadata PASSED [ 70%]
backend/tests/test_pysewer.py::test_pysewer_synthesize_topology PASSED [ 75%]
backend/tests/test_pysewer.py::test_real_elevation_differences_affect_slopes PASSED [ 79%]
backend/tests/test_routing.py::test_dry_routing_shortest_path PASSED [ 83%]
backend/tests/test_routing.py::test_flooded_nodes_avoided PASSED [ 87%]
backend/tests/test_routing.py::test_submerged_origin_returns_unreachable PASSED [ 91%]
backend/tests/test_routing.py::test_submerged_destination_returns_unreachable PASSED [ 95%]
backend/tests/test_routing.py::test_no_viable_path_all_blocked PASSED [100%]
===== 24 passed, 1 warning in 3.48s =====
```

6. Empirical Latency Benchmarks

Measured via `backend/tests/benchmark_api.py` across 30 iterations per endpoint on local hardware:

Endpoint	Method	Functionality	Mean Latency	Median	P95 Latency	Municipal Requirement
<code>/api/simulate</code>	POST	25-Node Inundation & Surge Solver	5.28 ms	5.23 ms	6.02 ms	< 50 ms (PASSED)
<code>/api/route</code>	POST	Dijkstra Safe Ambulance Routing	5.41 ms	5.38 ms	6.04 ms	< 50 ms (PASSED)
<code>/api/choke</code>	POST	Multi-Node Choke & Backwater Surge	8.22 ms	5.39 ms	25.43 ms	< 50 ms (PASSED)
<code>/api/pysewer/status</code>	GET	PySewer Methodology & Constraints	4.60 ms	4.62 ms	5.05 ms	< 50 ms (PASSED)
<code>/api/pysewer/synthesize</code>	POST	39-Conduit Pipe Layout & Sizing	7.41 ms	7.19 ms	9.04 ms	< 50 ms (PASSED)
<code>/api/network</code>	GET	GeoJSON Graph Topology	6.22 ms	6.20 ms	6.90 ms	< 50 ms (PASSED)

7. Final SIH-Safe Claim Audit

Claim in Technical Pitch	Safe in SIH?	Defensible Scientific Justification
"Real-time Open-Meteo rainfall"	SAFE	Genuinely calls live Open-Meteo REST API with fallback to local cache.
"PySewer gravity layout synthesis"	SAFE	Generates gravity DAG towards Minto outfall enforcing CPHEEO slope/velocity limits.
"Multi-node manhole choke simulation"	SAFE	Clicking multiple manholes zeroes capacity and dynamically propagates upstream surge.
"Dijkstra safe ambulance routing"	SAFE	NetworkX Dijkstra pathfinding prunes flooded roads >15 cm in <6 ms.
"Sub-second municipal response"	SAFE	Empirically verified at 5.28 ms mean simulation latency.
"Full 2D hydrodynamic solver (SWMM/Telemac)"	UNSAFE	Do not claim 2D hydrodynamics. Claim: "Physics-guided 1D Manning pipe conveyance with Rational overland sag cascade."
"Real municipal sewer blueprints"	UNSAFE	Do not claim real municipal sewer data. Claim: "Synthetic proxy drainage graph generated using PySewer methodology from OSM centerlines and DEM."
"Historical sensor validation (85% accuracy)"	UNSAFE	Do not claim validation against IoT sensors unless empirical sensor logs exist. Claim: "Topographically calibrated to historical Minto Bridge underpass sag blackspot."

8. 15-Step Judge Demonstration Sequence

- Launch App:** Open `http://localhost:5173`. Point out the dark glassmorphic dashboard, Minto Bridge centered map, and 6 top KPI cards.
- Step 1 — Dry Baseline:** Click 1. Dry in the demo bar. Rain is set to 0 mm/hr. Observe max flood depth is strictly 0.0 cm across all 25 nodes.
- Step 2 — Moderate Rain:** Click 2. Moderate. Rain is set to 35 mm/hr for 30 minutes. Minor pooling appears at Minto Underpass (6.4 cm, Low Risk, cyan).

4. **Step 3 — Monsoon Downpour:** Click 3. Downpour. Rain surges to 75 mm/hr. Point out Minto Bridge Underpass turning **CRITICAL RED** (>30 cm).
5. **Step 4 — Storm Duration Scaling:** Drag the Duration slider from 15 min to 60 min. Show cumulative volume scaling flood depth from 13.5 cm to 35.8 cm.
6. **Step 5 — Enable Choke Simulation:** Click 4. Choke or toggle the Choke switch. Blocked manhole halo rings appear on the map.
7. **Step 6 — Multi-Node Silt Clog:** Click additional nodes (e.g. ddu_marg_west). Observe multiple blocked halos.
8. **Step 7 — Backwater Surge:** Show upstream neighbors surcharging due to conduit blockage.
9. **Step 8 — Open Alert Center:** Click the Alert Center. Show severity filters (All / Critical / High) and individual alert dismissal.
10. **Step 9 — Drag & Minimize Alert Panel:** Drag the panel across the map by its header grip. Click – to collapse it into the floating glowing status pill. Click ↺ Reset to restore.
11. **Step 10 — Open Safe Ambulance Corridor:** Click 5. Route in the demo bar or toggle the Ambulance Route switch.
12. **Step 11 — Origin & Destination Callouts:** Point out the green ● **START: CP Outer Circle North** and red 🚒 **DESTINATION: Barakhamba Road** markers on the map.
13. **Step 12 — Flooded Segment Bypass:** Show how the green safe corridor dynamically diverts away from the submerged Minto Bridge underpass.
14. **Step 13 — Swap Direction:** Click ↕ Swap Direction in the sidebar. Dijkstra immediately recalculates the return evacuation corridor in <6 ms.
15. **Step 14 — PySewer Gravity Inspector:** Switch to the **PySewer** tab in the Alert Panel. Show the 39 synthesized conduits, CPHEEO self-cleansing velocity compliance ($v \geq 0.6$ m/s), slope percentages, and full-pipe flow capacities (Q_{cap}).
16. **Step 15 — Offline Resilience:** Disconnect Wi-Fi or stop the backend. The dashboard automatically switches to **OFFLINE DEMO MODE** and continues computing simulation and routing without interruption.